

A Cryptographic Infrastructure for Friend-to-Friend Choreographies

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A framework for

¹See [4, 5, 2, 1] for a formal model of local-first coordination.

A framework for

friend-to-friend [7, 8]

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A framework for

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coordination

and its (forthcoming) implementation

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*To appear on
JLAMP 2026*

- ▶ Motivations for re-considering distributed computing
(and not being "obscured by clouds")

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- ▶ A new choreographic model
 - ▶ F2F-choreographies
 - ▶ Nearbybytes

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- ▶ Motivations for re-considering distributed computing
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 - ▶ F2F-choreographies
 - ▶ Nearbytes
- ▶ Implementation aspects

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Blessing and curse of clouds

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- ▶ enabler of service-oriented computing/microservices (software-as-service)
- ▶ anytime-anywhere motto
- ▶ dynamic reconfiguration (services can be replaced if needed)
- ▶ reduced costs for maintenance, facilities, personnel, software, and hardware

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Where would you like to keep your data?
Can you control how they are used?

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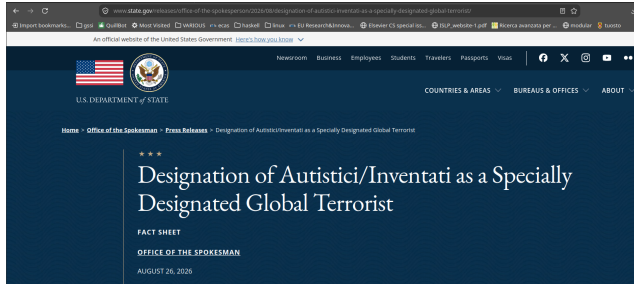
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Anytime, anywhere...really?

- ▶ *like during the AWS's outage on 25/11/2020*
- ▶ *or almost all Google services down on 14/12/2020*
- ▶ *DSL typical availability of 97%*
- ▶ *acutally, some SLA have no lower bound [What Is the Digital Divide?]*

...and limited autonomy/sovereignty



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en.wikipedia.org/wiki/Autistici/Inventati

QuillBot Most Visited VARIOUS ↗ ecas haskell linux ↗ EU Research&Innova... Elsevier CS special iss...

as Monuments: Photograph a monument, help Wikipedia and win!

more

Autistici/Inventati

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"A/I" redirects here. For other uses, see [AI \(disambiguation\)](#).

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Autistici/Inventati (A/I) was an Italian **hacktivist** group that provided Internet communication services for groups opposed to **fascism**, **militarism**, **racism**, **sexism**, **homophobia** and **transphobia**.^[1]

Abbreviation	A/I
Established	2010s
Dissolved	6 September 2026
Type	Non-profit
Services	Encryption, anonymity, and mapping of Internet events (2000s–2010s)

...and limited autonomy/sovereignty



en.wikipedia.org/wiki/Autistici/Inventati

Monuments: Photograph a monument, help Wikipedia and win!

Autistici/Inventati

Article Talk

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Perhaps some re-thinking is needed?

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 - ▶ computational resources

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- ▶ kept reliance on the cloud to a minimum?
- ▶ retained control over
 - ▶ data (i.e., ensure **confidentiality**)
 - ▶ computational resources
- ▶ had more "autonomy"

What if we go friend-to-friend too?

In Popescu, Crispo & Tanenbaum words

[...]Friends Team-Up and Beat the System

Abstract [...] while existing peer-to-peer architectures with similar aims attempt to build trust relationships on top of the basic, trust-agnostic, peer-to-peer overlay, Turtle takes the opposite approach, and builds its overlay on top of pre-existent trust relationships among its users. [...]

In Li & Dabek's words

F2F: reliable storage in open networks

Abstract [...] We argue that a node should choose its neighbors (the nodes with which it shares resources) based on existing social relationships instead of randomly. This approach provides incentives for nodes to cooperate and results in a more stable system which, in turn, reduces the cost of maintaining data. [...]

Local-first coordination
via
Friend-to-Friend Choreography

Our choreographic model

Syntax

$$\begin{aligned} F &::= A \rightsquigarrow B : t \\ &\quad | F + F' \\ &\quad | F ; F' \\ &\quad | F \mid F' \\ &\quad | \dots \end{aligned}$$

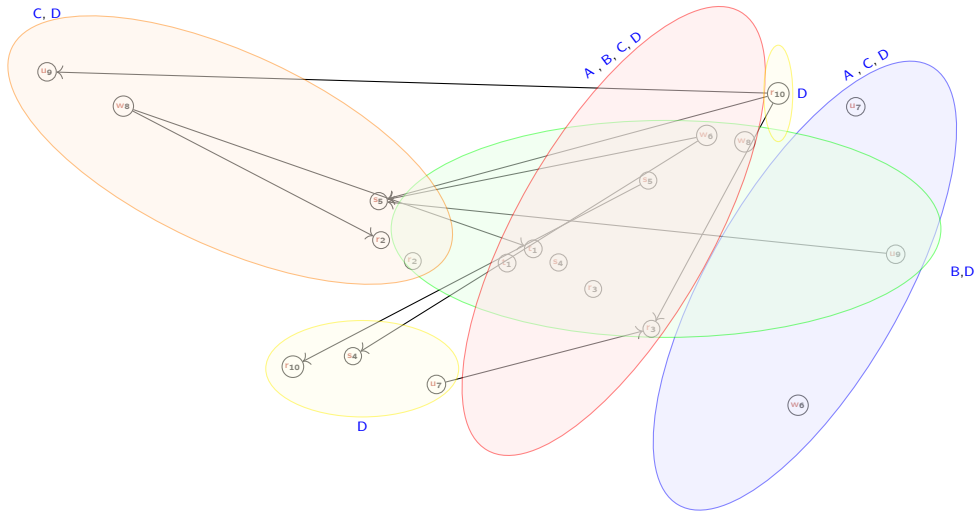
where

- ▶ B is a finite (possibly empty) set of A 's "buddies"
- ▶ t is a "type"

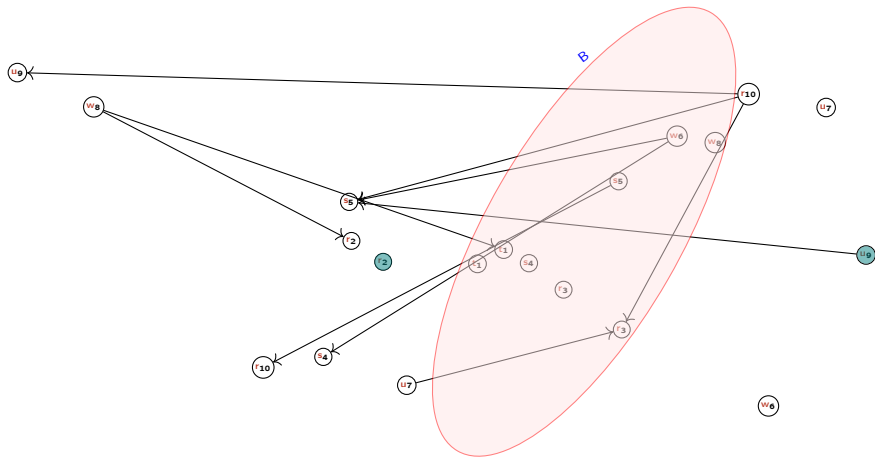
Semantics

$$\begin{aligned} \llbracket A \rightsquigarrow B : t \rrbracket &= \left\{ [A!B\langle t \rangle] \right\} \\ \llbracket F + F' \rrbracket &= \llbracket F \rrbracket \cup \llbracket F' \rrbracket \\ \llbracket F ; F' \rrbracket &= \{ r ;_{\varepsilon_r} r' \mid \dots \} \\ \llbracket F \mid F' \rrbracket &= \{ r ;_{\emptyset} r' \mid \dots \} \\ \llbracket \dots \rrbracket &= \dots \end{aligned}$$

The coordination model, informally



Replication



Having fun together

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Alice wants to arrange something for the evening with Bob, Carl, Daphne, and Evelin

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A's options are between staying home for a pizza or going out to a cinema

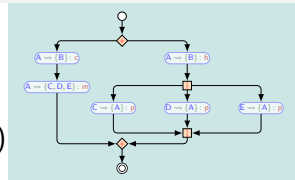
- ▶ if A opts for a cinema, B decides which movie; E should be oblivious of the title
- ▶ otherwise, deal is struck with whoever tells A which pizza they want

An F2F-choreography for fun

$$F = \begin{array}{l} A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p) \end{array}$$

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Let's spell out some requirements:

- ▶ h and c may be public, but let's keep the rest confidential
- ▶ E should be able to know if A decided to go out or B choose the movie
- ▶ C , D , and E cannot choose a pizza until A makes her choice
- ▶ X 's choice of pizza is known only to X and A

Nearbytes and its cryptographic infrastructure

What is Nearbybytes?

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- ▶ set of protocols for confidential interactions combining
 - ▶ communication via "physical wiring"
 - ▶ with cryptography + "hubs" merging

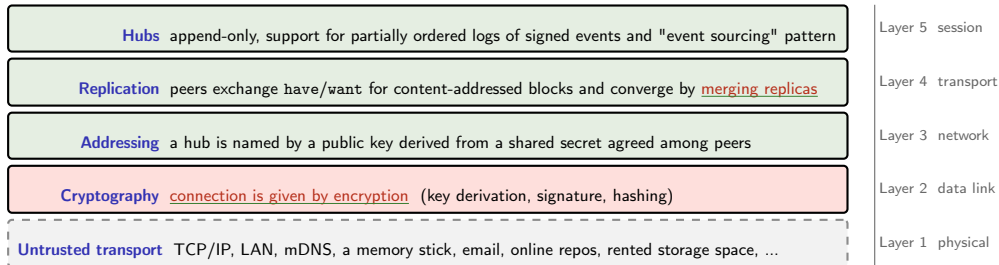
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- ▶ logical content-based addressing leveraging public-key cryptography
- ▶ data-related events form a partially ordered log

Nearbytes's stack

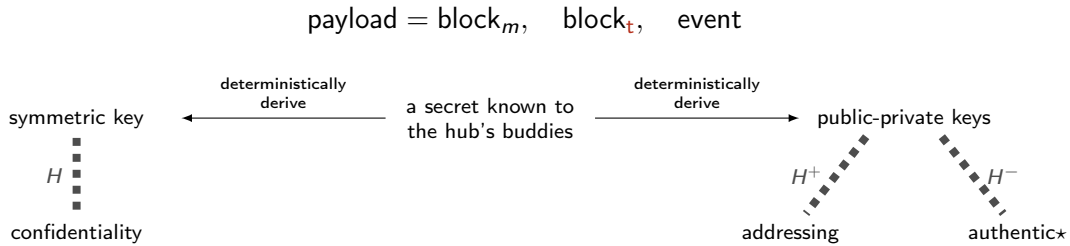


Nearbytes

Hush-hush (no key distribution)

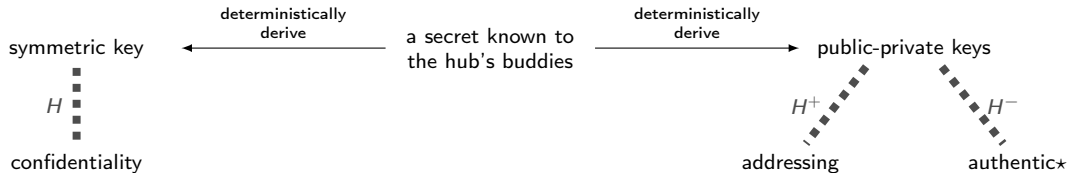
payload = block_m, block_t, event

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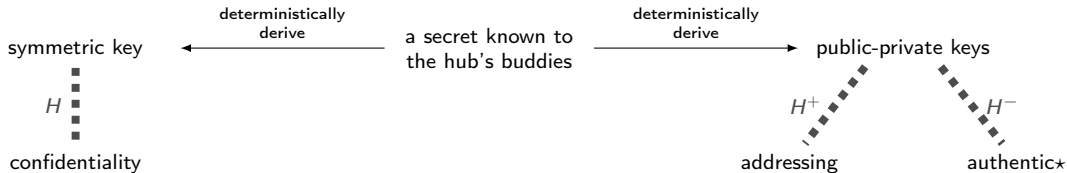
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$$\text{block}_m = \{ \text{msg} \}_H$$

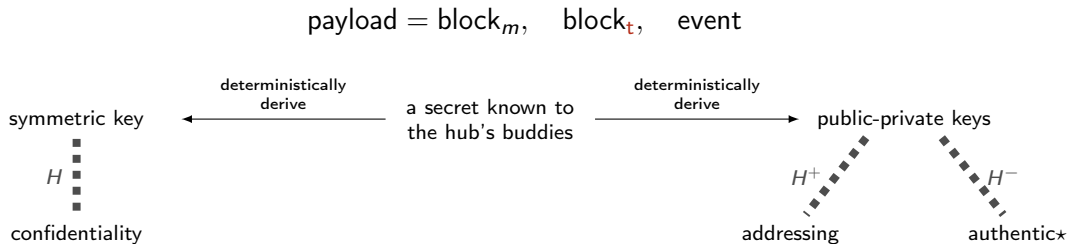
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block_m = { msg }_H
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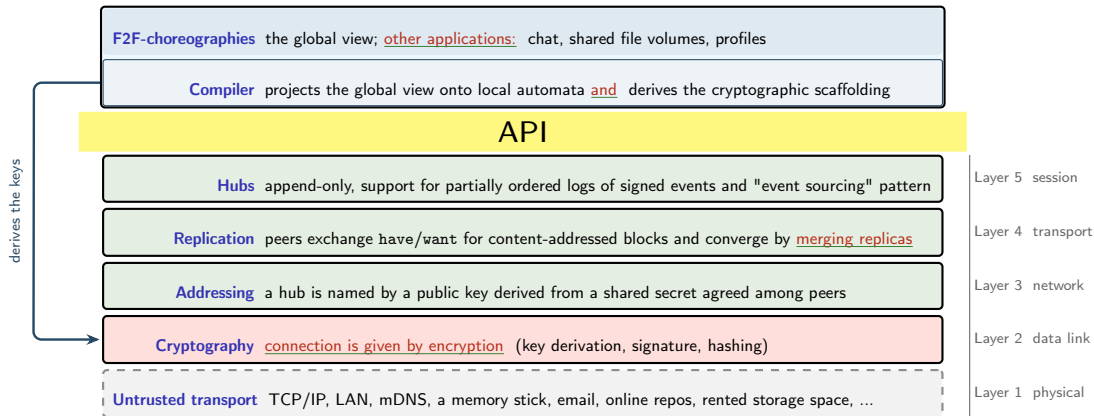
block_m = { msg }_H

block_t = { type }_{H⁺}

event = { hash(ev₁), ..., hash(ev_k), hash(block_m), hash(block_t) }_{H⁻}

F2F-choreographies & Nearbybytes

F2F-choreographies on top of Nearbybytes



Hubs' addressing

Buddies agree on a "master secret" M

Leverage buddies' public keys

$$\left\{ \begin{array}{lcl} H & = & M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \cdot E^+ \\ H_1 & = & M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \\ \dots & & \\ H_{14} & = & M \cdot A^+ \cdot B^+ \\ H_{15} & = & M \end{array} \right.$$

Projections

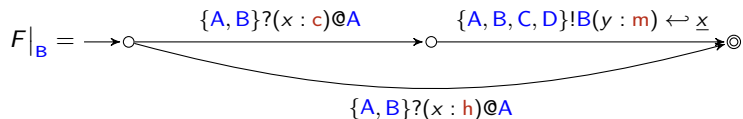
Deriving buddies' expected behaviour:

$$F = \quad A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p)$$

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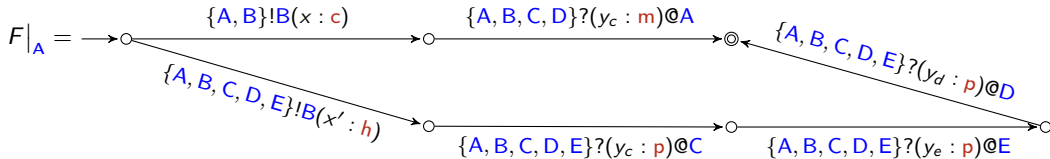
$$F = \begin{array}{l} A \rightsquigarrow B : \textcolor{blue}{c}; B \rightsquigarrow \{A, C, D\} : \textcolor{blue}{m} \\ + \quad A \rightsquigarrow B : \textcolor{blue}{h}; (C \rightsquigarrow A : \textcolor{blue}{p} \mid D \rightsquigarrow A : \textcolor{blue}{p} \mid E \rightsquigarrow A : \textcolor{blue}{p}) \end{array}$$



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$$F = \begin{array}{l} A \rightsquigarrow B : \textcolor{blue}{c}; B \rightsquigarrow \{A, C, D\} : \textcolor{red}{m} \\ + \quad A \rightsquigarrow B : \textcolor{blue}{h}; (C \rightsquigarrow A : \textcolor{red}{p} \mid D \rightsquigarrow A : \textcolor{red}{p} \mid E \rightsquigarrow A : \textcolor{red}{p}) \end{array}$$



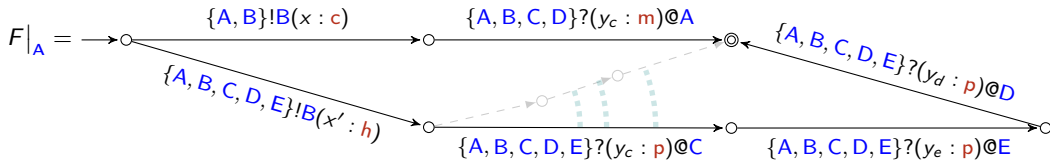
$$\left[A! \{A, B, C, D, E\} \langle \textcolor{blue}{h} \rangle \right]$$

$$e = \{ \textcolor{blue}{A} \text{'s address} \}_H, \{ \textcolor{blue}{h} \}_{H^+}, \{ \text{hash}(\{ \textcolor{blue}{A} \text{'s address} \}_H), \text{hash}(\{ \textcolor{blue}{h} \}_{H^+}) \}_{H^-}$$

Projections

Deriving buddies' expected behaviour:

$$F = \begin{aligned} & A \rightsquigarrow B : \textcolor{blue}{c}; B \rightsquigarrow \{A, C, D\} : \textcolor{brown}{m} \\ & + \quad A \rightsquigarrow B : \textcolor{blue}{h}; (C \rightsquigarrow A : \textcolor{brown}{p} \mid D \rightsquigarrow A : \textcolor{brown}{p} \mid E \rightsquigarrow A : \textcolor{brown}{p}) \end{aligned}$$



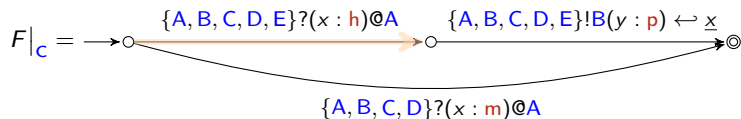
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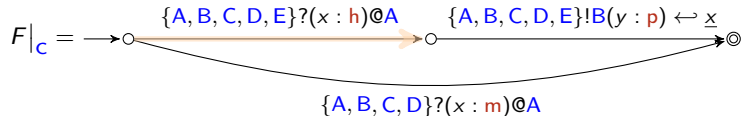
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$$\left[\begin{array}{c} A! \{A, B, C, D, E\} \langle \textcolor{blue}{h} \rangle \\ \downarrow \\ C! \{A, B, C, D, E\} \langle \textcolor{blue}{p} \rangle \end{array} \right]$$

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$$\{ \text{margherita} \}_H, \{ \textcolor{blue}{p} \}_{H^+}, \{ \text{hash}(e), \text{hash}(\{ \text{margherita} \}_H), \text{hash}(\{ \textcolor{blue}{p} \}_{H^+}) \}_{H^-}$$

Executing projections in Nearbybytes

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- ▶ Easy realisation: generate the automata and interpret them:
- ▶ More smartly: "on-demand" generation of the automata leveraging event-notification
- ▶ Compile the automata to actual code (e.g., Typescript or GoLang and use `Nearbybytes` as a library)

Future work

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- ▶ Study well-formedness conditions of F2F-choreographies for "odd" protocols like

$$A \rightsquigarrow m_1; A \rightsquigarrow m_2; B \rightsquigarrow m_3 + A \rightsquigarrow m_2; B \rightsquigarrow m_4 \quad (1)$$

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- ▶ Consider specific notions of eventual consistency
- ▶ Extend `Nearbytes` with broadcast encryption schemes?

Thank you!

On "Obscured by clouds" I like

Bridges burning gladly
Merging with the shadows,
Flickering between the lines.
Stolen moments floating softly on the air,
Borne on wings of fire
and climbing higher.

Ancient bonds are breaking,
Moving on and changing sides.
Dreaming of a new day,
Cast aside the other way.

Magic visions stirring,
Kindled by and burning flames rise in her eyes.

The doorway stands ajar,
The walls that once were high.
Beyond the gilded cage,
Beyond the reach of ties.
The moment is at hand.

She breaks the golden band.

(Pink Floyd)

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